

Log Analysis for Patterns

Log Data Analyzed:

Apr 10 10:10:12 Failed login for admin from 185.45.23.11 Apr 10 10:10:18 Failed login for admin from 185.45.23.11 Apr 10 10:10:25 Failed login for admin from 185.45.23.11 Apr 10 10:11:05 Successful login for admin from 185.45.23.11 Apr 10 10:11:45 New user account created: backdoor_user Apr 10 10:12:30 File accessed: /etc/passwd Apr 10 10:13:10 File accessed: /etc/shadow Apr 10 10:14:00 Large data transfer detected: 500MB sent to 185.45.23.11 Apr 10 10:15:22 Failed login for root from 192.168.5.44 Apr 10 10:15:28 Failed login for root from 192.168.5.44 Apr 10 10:15:35 Failed login for root from 192.168.5.44 Apr 10 10:15:50 Failed login for root from 192.168.5.44 Apr 10 10:16:10 Successful login for root from 192.168.5.44 Apr 10 10:17:00 sudo command executed by root Apr 10 10:18:00 Firewall rules modified by root Apr 10 10:19:00 Antivirus service stopped by root

Pattern 1: Brute Force Followed by Successful Login

The first clear pattern visible in the logs is the same brute force pattern we identified earlier. Three failed logins for admin from 185.45.23.11 within 13 seconds followed by a successful login at 10:11:05. Five minutes later a completely separate brute force attack happened targeting the root account from a different IP 192.168.5.44 with four failed attempts within 35 seconds followed by a successful login at 10:16:10. This confirms that the brute force activity is not isolated and is happening across multiple accounts from multiple IP addresses.

Pattern 2: Backdoor Account Creation

Exactly 40 seconds after the successful admin login at 10:11:05 a new user account called backdoor_user was created at 10:11:45. This is a classic attacker technique to maintain persistent access to the system even if the original admin password is changed. This activity was not performed by any known administrator and is a strong indicator of malicious post-compromise behavior.

Pattern 3: Sensitive File Access

At 10:12:30 and 10:13:10 the attacker accessed two extremely sensitive Linux system files. The first was /etc/passwd which contains a list of all user accounts on the system. The second was /etc/shadow which stores the hashed passwords of all user accounts. Accessing these files means the attacker was collecting credentials to crack offline and potentially compromise every user account on the system.

Pattern 4: Large Data Exfiltration

At 10:14:00 a massive data transfer of 500MB was detected being sent from the compromised system to the attacker's IP address 185.45.23.11. This is a clear sign of data exfiltration where the attacker is stealing large amounts of sensitive data and transferring it to their own server.

Pattern 5: Defense Evasion and System Tampering

After the root account was compromised at 10:16:10 the attacker executed sudo commands, modified firewall rules, and stopped the antivirus service. Modifying firewall rules means the attacker was opening backdoor ports for future access. Stopping the antivirus means they were preparing to install malware without being detected.

3. Threat Hunting Findings

Finding 1: Confirmed Brute Force Campaign Across Multiple Accounts

Our hypothesis was correct. The threat hunting exercise revealed that this was not a single isolated brute force attempt but a coordinated campaign targeting multiple privileged accounts. Both the admin account and the root account were targeted and successfully compromised within minutes of each other from two different IP addresses suggesting either a single attacker using multiple machines or a coordinated group attack.

Finding 2: Active Persistence Mechanism Installed

The creation of the backdoor_user account at 10:11:45 confirms that the attacker has established a persistence mechanism on the system. Even if all passwords are reset the attacker can still login using this hidden account. This account must be immediately identified and removed.

Finding 3: Credential Harvesting Detected

The access to /etc/passwd and /etc/shadow files confirms the attacker was attempting to harvest all user credentials from the system. If they successfully cracked the hashed passwords they now potentially have access to every user account on the system. All user passwords across the entire organization must be treated as compromised and reset immediately.

Finding 4: Data Breach Confirmed

The 500MB data transfer to IP 185.45.23.11 at 10:14:00 confirms that a data breach has occurred. The exact contents of the transferred data must be determined urgently. If customer data or financial records were included the organization may have legal obligations to notify affected individuals and regulatory authorities.

Finding 5: Security Controls Actively Disabled

The attacker modified firewall rules and stopped the antivirus service after gaining root access. This means the system's defenses have been actively dismantled from the inside. The system cannot be trusted in its current state and must be taken offline, fully audited, and rebuilt from a clean backup.